

Complete HyDRA pipeline description

Pipeline availability. Our pipeline integrates the currently available open-source tools in BASH scripts using basic UNIX commands. Our pipeline was designed to run in a high-performance computer (HPC) where computational tasks are allocated in a portable batch system (PBS). However, to make HyDRA straightforward, we have made it available in a range of different script options, including i) a Dockerfile to build a Docker container image with all tools/versions used in this study ii) step-by-step command lines files to guide users that wish to run the pipeline on a different system and or with the Docker container; and iii) the BASH scripts that write and submit PBS jobs. Additionally, for users without access to an infrastructure that supports Docker, the Docker container image itself has also been provided and can be converted into a Singularity/Apptainer container image, as described on the HyDRA GitHub repository. All these scripts, including the all supporting files for building the Docker container and a comprehensive README.md file, are available at the HyDRA's GitHub repository, <https://github.com/isabela42/HyDRA>, making the pipeline available to virtually any user.

Write to PBS scripts structure. HyDRA is structured in BASH scripts that write and submit portable batch system (PBS) jobs, meaning that our scripts were designed to run in a high-performance computer (HPC) where computational tasks, or simply jobs, are allocated in a PBS system. The in-house designed BASH scripts get arguments from command-line flags using getopt, which is a built-in Unix shell command for parsing options through the command-line, designed based on the C interface of getopt. Except for the summary-generating scripts, all others require the following flags: `-I` input TSV file containing the path to all input FASTQ/FASTA files, `-p` stem for PBS file names, `-e` email, `-m` memory, `-c` number of CPUS and `-w` clock wall time for the PBS jobs. HyDRA scripts produce log files and output folders with their execution time stamp, to allow the appropriate connection between input and output files and to prevent users from accidentally overwriting the generated files. For each input file provided in the TSV file, HyDRA's BASH scripts will write a PBS file with the following: i) a header session; ii) a PBS directives' session, containing the resources required to run each job according to provided flags; iii) a session to set the working directory appropriately; iv) a session to load the required PBS modules and/or locally installed tools that will be used in that step; v) a session with the actual command-lines to run the specific step; vi) a session to submit the PBS scripts that were just created by the BASH script; and vii) a final session to produced log files with all the input information.

Short-read treatment steps. HyDRA was designed based on a previous short-read-only approach published by us¹, incorporating its treatment steps for the short-read data. In the HyDRA pipeline, raw short-reads are corrected with Rcorrector v.1² and the uncorrectable reads are subsequently removed by FilterUncorrectablePEfastq.py script v.2016 (<https://github.com/harvardinformatics/TranscriptomeAssemblyTools>³). Illumina sequencing adaptors are removed by Trimmomatic v.0.36⁴ and then a user-defined database of ribosomal RNAs (rRNA) reads are filtered out using BBDuk v.39.01 (<https://sourceforge.net/projects/bbmap>⁵) – the same tool is also used to check the average and end quality of reads. Pre-processed reads are then mapped to the genome to flag any possible contamination using Bowtie2 v.2.2.9⁶ with a user supplied reference genome file and SAMtools v.1.9⁷. Quality control routines are performed with FastQC v.0.12.1 (<http://www.bioinformatics.babraham.ac.uk/projects/fastqc>⁸) and MultiQC v.1.14⁹ after each key processing step, to ensure quality standards are met or exceeded. The pipeline performs additional control steps on the pre-processed reads, i.e. strandness assessment of the RNAseq library is performed with RSeQC v.2.6.4¹⁰ to ensure strand-specificity, and replicates' correlation assessment is performed with DeepTools2 v.3.5.0¹¹.

Long-read treatment steps. First, raw long-reads are treated to remove adaptors with Porechop v.0.2.4 (<https://github.com/rrwick/Porechop>¹²), followed by trimming reads according to their average and head quality with Chopper v.0.5.0¹³, and the removal of poly(A) adaptor tails with Cutadapt v.3.9.13¹⁴. Next, to correct the long reads, we create a Burrows-Wheeler transform (BWT) from the short reads using RopeBWT2 v.r187¹⁵ and convert it to Run-Length Encoded (RLE)-BWT format using FMLRC2 v.0.1.7¹⁶, the top-performing hybrid error correction tool¹⁷. FMLRC2 v.0.1.7 is then used to correct the long-reads together with the produced NPY file. The downside of this tool being that corrected long-read sequences are outputted in a FASTA format, implying that base quality information is lost at this point. Next, a user-defined database of rRNA reads are filtered out using BBDuk v.39.01 – the same tool is also used to check the average and end quality of reads. Pre-processed reads are mapped to a user supplied reference genome file to flag contamination, using Bowtie2 v.2.2.9 and SAMtools v.1.9. Similarly to the short-read-only processing steps, quality control routines are performed with NanoPlot and NanoComp¹³ in addition to FastQC v.0.12.1 and MultiQC v.1.14 interspersed along the first two main processing steps. As part of our strict quality control, we have created a Python script to assess read length after each processing step, fasta_splitter.py. The pipeline also assesses pre-processed reads

regarding their RNAseq library strandness, using RSeQC v.2.6.4.

Hybrid *de novo* transcriptome assembly steps. Treated short- and long-reads are integrated to form a hybrid *de novo* transcriptome assembly. After an extensive search for state-of-the art tools that met our criteria, we selected rnaSPAdes v.3.14.1¹⁸ to combine the short- and long-read data into a hybrid *de novo* transcriptome assembly. Splicing detection and read representation were performed with GMAP v.2023-07-20¹⁹ and Bowtie2 v.2.2.9. We removed highly redundant transcripts (i.e. 98% identity and >90% coverage) using CD-HIT v.4.6.8²⁰. Transcripts were then differentiated from sequencing noise using reads per kilobase per million (RPKM) values obtained based on Minimap2 v.2.26²¹ and Bowtie2 v.2.2.9 alignments, for long and short reads respectively. RPKM read counts were then estimated using SAMtools idxstats report^{6,7,21}. The RPKM values were calculated as described by the NIH Bioinformatics Training and Education Program (<https://bioinformatics.ccr.cancer.gov/btep/questions/what-is-the-difference-between-rpkm-fpkm-and-tpm>), where sequencing depth was normalized by dividing read counts by reads per million and further normalized by transcript lengths. We decided HyDRA would not use unmapped reads to calculate RPKM, since only aligned reads are usually considered assessing read support (for more details see <https://www.biostars.org/p/278329>²²). HyDRA runs with two user-defined thresholds sets of read support, one for monoexonic/unspliced transcripts, and another for multiexonic/spliced transcripts. Additionally, HyDRA assesses completeness, accuracy and outputs overall assembly metrics. For that, all sets of transcripts are aligned to a user-defined GFF reference genome with GMAP v.2023-07-20 to create GFF format annotation files. The resulting assembly has its quality assessed using BUSCO v.20161119²³, Trinity Stats v.2.8.4²⁴ and TransRate v.1.0.3²⁵. BUSCO runs based on Augustus 3.2.2 and odb9 together with the eukaryotic, vertebrata and mammalian ortholog sets to assess assembly completeness²³. TransRate and Trinity Stats score assemblies based on experimental evidence (e.g. the sequenced reads)^{24,25}.

LncRNA discovery and annotation steps. LncRNAs discovery is performed using four machine learning models and selecting consistently predicted transcripts. Three machine learning models from ezLncPred v.1.0²⁶, i.e. CPAT, CNCI, CPC2, are used to predict the coding potential of transcripts. To identify candidate lncRNAs, we use FEELnc v.0.2²⁷. The user must supply the pipeline with a database of annotated protein-coding genes and of confirmed lncRNA genes to train the machine learning algorithm of FEELncfilter - here we used the GENCODE GRCh38 annotated protein-coding and a database of confirmed lncRNA genes¹, which filtered

non-lncRNA transcripts and transcripts shorter than 200 nucleotides. The FEELncodpot module was used to compute a coding potential score and assign a confidence level to potential ORFs. The FEELncclassifier module was then used to detect the nearest annotated transcript and further classify the lncRNAs based on genomic position relative to protein-coding genes. We considered all transcripts with no coding potential detected by at least two different ezLncPred tools and FEELnc candidate lncRNAs to be true positives.

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Running HyDRA with normal ovarian and fallopian tube data

Raw reads quality. Short- and long-read raw data was above the expected minimum quality thresholds. Quality control (QC) routines were implemented throughout the entire pipeline to ensure the high quality of the input data for *de novo* assembly^{1,2}, with the majority of these routines performed after every read processing step. This was implemented to confirm that our processing and correction of reads was not detrimental to overall quality. Instead, we expect quality values to be maintained or increase after treatment. To assess long-read data quality, it is common practice to use NanoPack³, rather than FastQC/MultiQC (<http://www.bioinformatics.babraham.ac.uk/projects/fastqc>⁴ and ⁵). Since the overall quality of short reads is expected to be much higher, the thresholds implemented on FastQC are not optimal for assessing long reads, i.e. quality must be compared against expected values and these are much lower for long-read data. For comparison, our raw data included nearly 10% more reads above Q10 than a recent study showing R10.4 (used here) to generate better quality reads compared to an older ONT flow cell⁶. However, this is still extremely low quality compared to that of short-reads.

Short-read correction. Less than 1% of bases were corrected in each sample using Rcorrector (average of 0.66%; Table S2)⁷ with 7.70-11.30% of the read pairs being removed per sample with the FilterUncorrectablePEfastq.py script (<https://github.com/harvardinformatics/TranscriptomeAssemblyTools>⁸). Following Bitar *et al.*⁹ recommendations for trimming Illumina adaptors, an average of 86.25% of read pairs survived trimming with Trimmomatic¹⁰ and an additional average of 9.66% and 2.35% forward and reverse reads survived independently (Table S2). From these, only 1.17% of the reads were below the minimum average read quality threshold of Q20 after trimming.

Long-read correction. In HyDRA, we used the best performing hybrid error correction tool, according to a recent benchmarking study¹¹: FMLRC2 uses a BWT built from highly accurate short-reads to correct erroneous subregions of error-prone long reads¹². A downside of this tool is that it only outputs FASTA files. Therefore, we first trimmed long reads to evaluate their quality improvement and then proceeded to correct them. Adaptors were removed with Porechop (<https://github.com/rwwick/Porechop>¹³), reads with either an average or beginning of the sequence quality below the minimum Phred score of Q10 were removed using Chopper³, and poly(A) tails were trimmed using Cutadapt¹⁴. These tools are widely used by the community^{15,16}.

Over 99% of the reads had adaptors removed from the start of the sequence and 96% of these passed the quality threshold (Table S2).

Quality assessment of treated reads. To rule out external contaminants, both short- and long-pre-processed reads were aligned to the most recent GENCODE version of the human genome¹⁷. Overall alignment rates varied from 79.72% to 85.01% per short-read sample (average 83.58%; Table S2), which falls within the expected range for high-quality RNAseq experiments¹⁸. For long reads, however, the alignment rate is on average 33.50% when using Bowtie2 (Table S2), mostly because this tool is not appropriate for long-read alignments¹⁹ (further details at <http://complex.zesoi.fer.hr/en/blog-en/56-gmap-vs-minimap2>). We also confirmed strandness of the sequenced reads using the infer_experiment.py routine from RSeQC²⁰, confirming that an average of 96.66% of the reads were explained by “1+,-,1-+,2++,2”, which stands for forward read as first strand (Table S2). Using the multiBamSummary, plotCorrelation and plotPCA functions of DeepTools²¹, we generated scatter plots of pairwise Pearson correlations and a clustered heatmap of sample correlations (Figure A). These analyses revealed an overall absence of batch effects, with FTSEC and OSEC samples forming independent clusters with the exception of FC-0081-2839, which was the sample with the lowest ng/ μ l concentration in the RNAseq library. All sample correlations (measured by Pearson coefficients) were higher than 0.84 (Table S2).

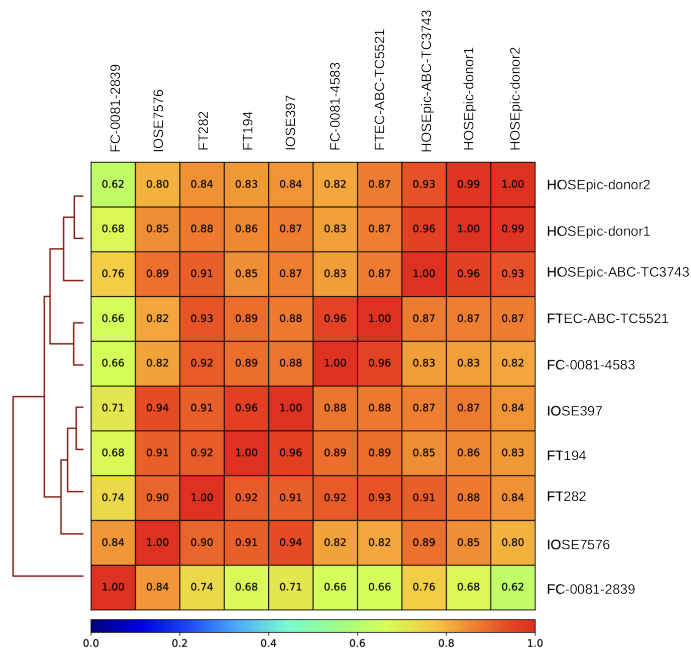


Figure A. Heatmap of short-read samples correlations. Plots were based on pairwise Pearson correlations, both generated with DeepTools.

Chimeric transcripts. In HyDRA, chimeric transcripts are kept together with the spliced isoforms for further analysis. During the splicing assessment described above, we observed the presence of chimeric transcripts in all assemblies. These would normally be filtered out, but these are relevant in the context of cancer, where chimeric transcripts are known to be formed through gene fusions resulting from e.g. chromosomal rearrangements²². Others have shown these transcripts to be of functional relevance to the phenotypes of cancer emergence and progression^{23,24,22,25}. We therefore decided these transcripts should be maintained in our hybrid assembly, for further studies. Careful examination of the CIGAR strings after alignment to the reference genome revealed that certain chimeric transcripts were being categorized as spliced, unspliced, or a combination of both, which we treated as a technical artifact of the alignment tool. In the HyDRA pipeline, we introduced a step to uniformly classify chimeric transcripts as spliced due to their inherent biological structure. We do so by reclassifying the chimeric alignments as spliced-alignments.

Benchmarking of tools for read correction and assembly. We selected tools for HyDRA based on previous studies where authors assessed multiple tools for specific methods (e.g. ¹¹ and ²⁶). However, we also performed in-house benchmarking of crucial steps, including custom transcriptome assembly and long-read error correction methods, which we describe below.

Assemblers. We benchmarked different assembly methods, including short-read-only, long-read-only and hybrid assemblers. Our findings support rnaSPAdes as the best-performing hybrid assembler, as previously shown by Zhang and collaborators¹¹. In summary, we designed our study to include two assemblers for each modality: short-read-only (Trinity and StringTie2), long-read-only (StringTie2 and UNAGI) and hybrid *de novo* transcriptome assembly (rnaSPAdes and StringTie2). We initially attempted to test all assemblers using raw sequencing reads, to remove any benefits from quality control steps implemented in the HyDRA pipeline. However, each tool presented a specific challenge: (1) Trinity requires a specific structure for sequence identifiers - for which we created an in-house script, used in HyDRA; (2) StringTie2 ran successfully on raw reads in short-read-only and long-read-only modes, however failed in hybrid mode; (3) UNAGI ran successfully, although nearly 700 GB of memory (695268796kb) and over 10 days (256:19:27) of processing were needed; (4) rnaSPAdes can only assemble adapter-free reads - running out of memory in three separate attempts to override this requirement. We therefore decided to use HyDRA-treated reads to test all tools, standardizing input and in theory allowing every tool to perform. Unfortunately, UNAGI requires a FASTQ file as input and our long-read

correction method only output FASTA files, which cannot be converted. Therefore, we had to remove UNAGI from the experiment. We evaluated the performance of all other tools in different assembly modalities using the same adaptation of the strategy from Hölzer and Marz (2019)²⁷ as described in the main manuscript. Briefly, a subset of the quality assessment metrics described by Hölzer and Marz is converted to a normalized score (0-1) from which a summarized score is calculated to rank all tools by performance. In our experiments, rnaSPAdes continues to show the best summarized score (4.48), a >35% improvement compared to Trinity (3.27), and below all three StringTie2 strategies (all <3.00 with short-read-only: 2.06; long-read-only: 2.81; hybrid: 2.45).

Long-read correction tools. We initially enlisted several long-read correction methods and intended to test all, namely: ECTools, FMLRC2, HALC, HG-CoLoR, Jabba and LoRDEC. However, running most of these methods posed significant challenges. ECTools and HALC require short reads to be pre-assembled before they can be used to correct long reads, which we reasoned defeats our purposes of testing correction methods (as we would not be able to uncouple the correction tool performance from that of the underlying assembler). HG-CoLoR has unique requirements for input files, where all short reads should be concatenated in just one file (even paired reads need to be separated into singles to ensure just one input file) and all long reads in one other file. Jabba has a similar problem to ECTools and HALC. While it does not require a pre-assembly per se, it does need a pre-built de Bruijn graph. We then compared FMLRC2, HG-CoLoR and LoRDEC, correcting long reads and aligning the results to the reference human transcriptome. HG-CoLoR behaved unexpectedly, creating corrected fragments but not reporting long reads. We explored possible causes for this issue, worth noting we attempted to use different k-mer lengths and allow the program to use more resources. None of these were successful. Finally, we were left with FMLRC2 and LoRDEC, which resulted in respective error rates of 2.011285e-01 and 2.012423e-01, thus confirming FMLRC2 as the best-performing tool, even though the difference is arguably non-significant.

In summary, our benchmarking experiments confirm the tools previously reported as best-performing for each task and support the choices made for HyDRA.

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